CONSTANTS OF FUNCTIONAL EQUATIONS OF L-FUNCTIONS

3.8–4.7

$$\begin{array}{ccc} W(\overline{K}/L) & \longrightarrow & W(\bar{\ell}/\ell) \\ \big\downarrow & & \big\downarrow \\ W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k). \end{array}$$

This reduces (3.8.2) to the following well-known lemma.

Lemma 3.9. Let F be an endomorphism of V, let n be an integer, and let F_n be the endomorphism of

$$V_n = V \otimes \mathbb{C}^n$$

such that, for a basis $e_0, \dots, e_{n-1}$ of $\mathbb{C}^n$,

$$F_n(v \otimes e_i) = v \otimes e_{i+1} \quad (0 \leq i < n-1), \qquad F_n(v \otimes e_{n-1}) = F(v) \otimes e_0.$$

Then

$$\det(1 - F_n t) = \det(1 - F t^n).$$

By the principle of continuation of algebraic identities, one may suppose F diagonal in a suitable basis of V. One is immediately reduced to the case $\dim(V) = 1$, where F is multiplication by a number a^n, $a \neq 0$. In the basis $f_i = a^{-i}(v \otimes e_i)$, one has $F_n(f_i) = a f_{i+1}$, for $i \in \mathbb{Z}/n\mathbb{Z}$. The eigenvectors of F_n are the

$$\sum_i \chi(i) f_i$$

where χ runs through the characters of $\mathbb{Z}/n\mathbb{Z}$, and the determinant identity follows.

3.10. Let K be a global field, an $\mathfrak{A}$-field in the sense of [15]. For every place v of K, write K_v for the local field which is the completion of K at v. The objects defined above for local fields will be denoted, for K_v, with a subscript v. Let $\mathbb{A}$ be the adele ring of K, let $\| \ \|$ be the absolute value satisfying

$$d(ax) = \|a\| \, dx$$

for a Haar measure dx on $\mathbb{A}$, and let ω_s be the quasi-character $\|x\|^s$ of $\mathbb{A}^*$. If x_v is the component of $x \in \mathbb{A}$ in K_v, then

$$\|x\| = \prod_v \|x_v\|_v.$$

Let dx denote the unique Haar measure on $\mathbb{A}$ such that

$$\int_{\mathbb{A}/K} dx = 1$$

(the Tamagawa measure), let d^*x be a Haar measure on $\mathbb{A}^*$, and let ψ be a character trivial on $\mathbb{A}/K$, with local components ψ_v. For f a C^∞ function with compact support on $\mathbb{A}$, put

$$\widehat{f}(y) = \int_{\mathbb{A}} f(x)\psi(xy) \, dx.$$

3.11. Tate's global functional equation [14] says, for χ a quasi-character of $\mathbb{A}^*/K^*$,

$$\int_{\mathbb{A}^*} \widehat{f}(x)\chi'(x)\,d^*x = \int_{\mathbb{A}^*} f(x)\chi(x)\,d^*x, \qquad \chi' = \omega_1\chi^{-1}, \tag{3.11.1}$$

the integrals being defined by analytic continuation in the families $\chi\omega_s$. Put

$$L(\chi) = \prod_v L(\chi_v) \tag{3.11.2}$$

by analytic continuation, and, for $dx = \otimes_v dx_v$,

$$\varepsilon(\chi) = \prod_v \varepsilon(\chi_v, \psi_v, dx_v). \tag{3.11.3}$$

The product (3.11.3) is independent of the choice of ψ and of the decomposition of dx, as follows from (3.3.2) and (3.3.3). If, for almost all v,

$$\int_{\mathcal{O}_v} dx_v = 1,$$

then almost all its factors are equal to 1. Formulas (3.3.1) and (3.11.1) give the global functional equation of Hecke L-functions:

$$\varepsilon(\chi)L(\chi') = L(\chi). \tag{3.11.4}$$

3.12. The results stated below are proved in [1] and [19]. Let K be a global field, $\overline{K}$ an algebraic closure of K, and V a complex representation of $\mathrm{Gal}(\overline{K}/K)$, always assumed continuous: the action of $\mathrm{Gal}(\overline{K}/K)$ factors through a finite Galois group $\mathrm{Gal}(K'/K)$. For every place v of K, choose a place of $\overline{K}$ above it, and let V_v be the representation obtained from V by restriction to the subgroup

$$W(\overline{K}_v/K_v)$$

of the decomposition group $\mathrm{Gal}(\overline{K}_v/K_v)$.

(A) For χ a quasi-character of $\mathbb{A}^*/K^*$, with local components χ_v, the infinite product

$$L(V,\chi) = \prod_v L(V_v \otimes \chi_v)$$

can be defined by analytic continuation in χ in the families $\chi\omega_s$. It converges for $\Re(s)$ large and extends to a meromorphic function of s.

(B) Let V^* denote the dual of V. One has

$$L(V,\chi) = \varepsilon(V,\chi)\,L(V^*,\chi'), \qquad \chi' = \omega_1\chi^{-1},$$

with $\varepsilon(V,\chi) \in \mathbb{C}^*$, of the form Ae^{Bs} as χ varies in the families $\chi\omega_s$.

(C) Let L be a finite extension of K in $\overline{K}$, let $N_{L/K}$ be the norm, let V_L be a representation of $\mathrm{Gal}(\overline{K}/L)$, and put

$$V_K = \mathrm{Ind}_{\mathrm{Gal}(\overline{K}/L)}^{\mathrm{Gal}(\overline{K}/K)}(V_L).$$

For χ a quasi-character of $\mathbb{A}_K^*/K^*$, one has

$$L(V_K,\chi) = L(V_L, \chi\circ N_{L/K}),$$

and similarly

$$\varepsilon(V_K,\chi) = \varepsilon(V_L, \chi\circ N_{L/K}).$$

(D) One also has

$$\varepsilon(V'\oplus V'',\chi) = \varepsilon(V',\chi)\varepsilon(V'',\chi),$$

which makes it possible to define $\varepsilon(V,\chi)$ when V is only a virtual representation.

The assertions (A) and (B) are proved by reduction to the case $\dim V = 1$, via Brauer's theorem. Assertion (C) follows from (B) and from the compatibility of norm and transfer recalled above.

4. Existence of local constants

Theorem 4.1. There exists a unique function ε, satisfying conditions (1) to (4) below, which associates a number

$$\varepsilon(V, \psi, dx) \in \mathbb{C}^*$$

to each isomorphism class of sextuples $(K, \overline{K}, \psi, dx, V, \rho)$, consisting of a local field K, an algebraic closure $\overline{K}$ of K, a nontrivial additive character $\psi : K \to \mathbb{C}^*$, a Haar measure dx on K, a finite-dimensional complex vector space V, and a representation

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V).$$

(1) For every exact sequence of representations

$$0 \to V' \to V \to V'' \to 0,$$

one has

$$\varepsilon(V, \psi, dx) = \varepsilon(V', \psi, dx)\varepsilon(V'', \psi, dx).$$

This condition shows that $\varepsilon(V, \psi, dx)$ depends only on the class of V in the Grothendieck group of representations of $W(\overline{K}/K)$, and allows $\varepsilon(V, \psi, dx)$ to be interpreted when V is only a virtual representation.

(2)

$$\varepsilon(V, \psi, a\,dx) = a^{\dim V}\varepsilon(V, \psi, dx).$$

In particular, for virtual V of dimension 0, $\varepsilon(V, \psi, dx)$ is independent of dx; it is denoted $\varepsilon(V, \psi)$.

(3) If L is a finite separable extension of K in $\overline{K}$, and V_K is the virtual representation of $W(\overline{K}/K)$ induced by a virtual representation of dimension zero V_L of $W(\overline{K}/L)$, then

$$\varepsilon(V_K, \psi) = \varepsilon(V_L, \psi \circ \mathrm{Tr}_{L/K}).$$

(4) If $\dim V = 1$, and ρ is defined by a quasi-character χ of K^*, then

$$\varepsilon(V, \psi, dx) = \varepsilon(\chi, \psi, dx).$$

4.2. Let

$$v = \sum_{i,j} n_{ij}(H_i, \chi_{ij}) \in R_0^+(W(\overline{K}/K))$$

(1.10). Assume that the H_i are pairwise nonconjugate. Let K_i be the extension of K defined by H_i, and again denote by χ_{ij} the quasi-character of K_i^* defined by χ_{ij}. For any choice of Haar measures dx_i on the K_i, and for ψ an additive character of K, put

$$\varepsilon(v, \psi) = \prod_{i,j} \varepsilon(\chi_{ij}, \psi \circ \mathrm{Tr}_{K_i/K}, dx_i)^{n_{ij}}. \tag{4.2.1}$$

It follows from (3.3.2) that $\varepsilon(v, \psi)$ is independent of the choice of the dx_i.

Lemma 4.3. With the notation 1.9.8, for $v \in R_0^+(W(\overline{K}/K))$,

$$\varepsilon(v, \psi(ax)) = \det(v)(a)\,\varepsilon(v, \psi).$$

It is enough to check this for $v = (H, \chi) - (H, 1)$, where H corresponds to an extension L of K. Let v_0 be the virtual representation $[\chi] - [1]$ of $W(\overline{K}/L)$. Formula (3.3.3) gives the assertion over L. It is known ([10], XII, §4) that the inclusion of K^* into L^* is identified with the transfer

$$W(\overline{K}/K)^{\mathrm{ab}} \longrightarrow W(\overline{K}/L)^{\mathrm{ab}}.$$

Since $b(v) = \mathrm{Ind}(v_0)$, the lemma follows from 1.2.

Lemma 4.4. The theorem is true in the archimedean case.

For $K \simeq \mathbb{C}$, define ε by (1) and (4), and (2) is automatic. For $K \simeq \mathbb{R}$, one checks from 2.9 that

$$b : R^+(W(\overline{K}/K)) \longrightarrow R(W(\overline{K}/K))$$

is surjective, with kernel generated by the $\chi_s - \chi_t$ where

$$\chi_s = (K^*, \omega_s) - [\omega_s] - [\chi^{-1}\omega_{s+1}].$$

For a virtual representation V, and $v \in R_0^+(W(\overline{K}/K))$ such that

$$b(v) = V - \dim(V)[1],$$

put

$$\varepsilon(V, \psi, dx) = \varepsilon(v, \psi)\, \varepsilon(1, \psi, dx)^{\dim V}. \tag{4.4.1}$$

We check that the right-hand side is independent of the choice of v, i.e. that, for v satisfying $b(v) = 0$, one has $\varepsilon(v, \psi) = 1$. By 4.3, it is enough to check this for $\psi(x) = \exp(2\pi i x)$. For this choice of ψ, the assertion follows from the preceding determination of $\mathrm{Ker}(b)$, and from the fact that the right-hand sides of (3.4.1), (3.4.2) are independent of s. It is clear that ε defined by (4.4.1) satisfies (1) to (4).

We now restrict to the nonarchimedean case.

4.5. Let K be a nonarchimedean local field, $\overline{K}$ a separable closure of K, and $K_{nr} \subset \overline{K}$ the maximal unramified extension of K. Let J be an open normal subgroup of the inertia group

$$I = \mathrm{Gal}(\overline{K}/K_{nr})$$

and let L be the corresponding extension of K_{nr}. For $\sigma \in I/J$, put

$$t_{I/J}(\sigma) = \inf\bigl(v_L(\sigma(x) - x) \mid x \text{ integral in } L\bigr),$$

and define functions $a_{I/J}$ and $sw_{I/J}$ by

$$\begin{aligned} a_{I/J}(\sigma) &= -t_{I/J}(\sigma) \quad (\sigma \neq e), \\ sw_{I/J}(\sigma) &= 1 - t_{I/J}(\sigma) \quad (\sigma \neq e), \\ \sum_\sigma a_{I/J}(\sigma) &= \sum_\sigma sw_{I/J}(\sigma) = 0. \end{aligned} \tag{4.5.1}$$

By Artin (see [10], VI, §2, th. 1), $a_{I/J}$ is the character of a representation $A_{I/J}$ of I/J, the Artin representation. By loc. cit., prop. 2, $sw_{I/J}$ is likewise the character of a representation $Sw_{I/J}$, the Swan representation. These representations are defined only up to isomorphism. $A_{I/J}$ is the sum of $Sw_{I/J}$ and the augmentation representation of I/J. For $I \supset J \supset K$, one has (loc. cit., prop. 3)

$$(A_{I/K})^{J/K} \xrightarrow{\sim} A_{I/J}, \qquad (Sw_{I/K})^{J/K} \xrightarrow{\sim} Sw_{I/J}. \tag{4.5.2}$$

If d is the valuation of the discriminant $\delta_{L/K_{nr}}$, i.e. the valuation of the different $\mathfrak{D}_{L/K}$ ([10], III, §3, prop. 6), and r denotes a regular representation, then (loc. cit., prop. 4)

$$A_{I/K}|_{J/K} = d\, r_{J/K} + A_{J/K}. \tag{4.5.3}$$

Let V be a representation of $W(\overline{K}/K)$, trivial on $J \subset I$, with character ψ. Define the Artin and Swan conductors of V by

$$\begin{aligned} a(V) &= \dim \mathrm{Hom}_{I/J}(A_{I/J}, V) = sw(V) + \dim V - \dim V^I, \\ sw(V) &= \dim \mathrm{Hom}_{I/J}(Sw_{I/J}, V) = \frac{1}{|I/J|} \sum_\sigma sw_{I/J}(\sigma)\psi(\sigma). \end{aligned} \tag{4.5.4}$$

By 4.5.2, these conductors are independent of the choice of J. For V of dimension 1, defined by a quasi-character χ, one has (loc. cit., prop. 5)

$$a(V) = \text{the conductor } a(\chi) \text{ of } \chi. \tag{4.5.5}$$

The preceding notation is therefore compatible with that introduced in 3.4.

With the notation 1.9.8, one has:

Lemma 4.5. For $v \in R_0^+(W(\overline{K}/K))$ and χ an unramified quasi-character of K^*,

$$\varepsilon(v \cdot \chi, \psi) = \chi(\pi)^{a(v)} \varepsilon(v, \psi).$$

It suffices to verify this for $v = (H, \lambda) - (H, \mu)$, where H defines an extension L of K, and λ, μ are two quasi-characters of L^*. Let d be the valuation of the different of L/K, let f be the residue-degree, let $I \subset W(\overline{K}/K)$ and $J \subset W(\overline{K}/L)$ be the inertia groups, and choose $K \subset J$ small enough. We compute $a(v)$. The adjunction formula for induced representations, (4.5.3), and (4.5.5) give, for every quasi-character ν of H, i.e. of L^*,

$$a((H,\nu)) = \dim \mathrm{Hom}_{I/K}(A_{I/K}, \mathrm{Ind}_H^W(\nu)) = f \, \dim \mathrm{Hom}_{J/K}(A_{J/K} + d\, r_{J/K}, [\nu]|_J) = f\, a(\nu) + fd. \tag{4.5.1'}$$

Thus

$$a(v) = f(a(\lambda) - a(\mu)).$$

For ν a quasi-character of L^*, χ_1 an unramified quasi-character of L^*, and $n = n(\psi \circ \mathrm{Tr}_{L/K})$ as in 3.4, one has by (3.4.3.1) or (3.4.3.2)

$$\frac{\varepsilon(\nu\chi_1, \psi \circ \mathrm{Tr}, dx)}{\varepsilon(\nu, \psi \circ \mathrm{Tr}, dx)} = \chi_1(\pi_L)^{a(\nu)+n}.$$

Hence

$$\varepsilon(v \cdot \chi, \psi)\varepsilon(v, \psi)^{-1} = \chi \circ N_{L/K}\left(\pi_L^{a(\lambda)-a(\mu)}\right) = \chi(\pi)^{f(a(\lambda)-a(\mu))},$$

which is the announced formula.

Terminology 4.6. Let K_1 be a Galois extension, finite or not, of K, with Galois group G. Let α be a quasi-character of K^*, and ψ a nontrivial additive character of K. We shall say that there exists an α, ψ-theory of constants for K_1/K if there exists $\varepsilon_{\alpha,\psi}$ of the following type:

(0) For H an open subgroup of G, defining a finite extension L of K, for V a representation of H, and dx a Haar measure on L,

$$\varepsilon_{\alpha,\psi}(V, dx) \in \mathbb{C}^*$$

is defined.

(1) For every exact sequence of representations

$$0 \to V' \to V \to V'' \to 0,$$

one has

$$\varepsilon_{\alpha,\psi}(V, dx) = \varepsilon_{\alpha,\psi}(V', dx)\varepsilon_{\alpha,\psi}(V'', dx).$$

Thus $\varepsilon_{\alpha,\psi}$ still makes sense for virtual representations.

(2) One has

$$\varepsilon_{\alpha,\psi}(V, a\, dx) = a^{\dim V} \varepsilon_{\alpha,\psi}(V, dx).$$

In particular, for virtual V of dimension 0, $\varepsilon_{\alpha,\psi}(V, dx)$ is independent of dx; it is denoted $\varepsilon_{\alpha,\psi}(V)$.

(3) For $H_1 \subset H_2 \subset G$, defining $L_1 \supset L_2$, and V a virtual representation of dimension 0 of H_1, one has

$$\varepsilon_{\alpha,\psi}(\mathrm{Ind}_{H_1}^{H_2} V) = \varepsilon_{\alpha,\psi}(V).$$

(4) For V of dimension 1, defined by a character χ of L^*,

$$\varepsilon_{\alpha,\psi}(V, dx) = \varepsilon(\chi\, \alpha \circ N_{L/K},\, \psi \circ \mathrm{Tr}_{L/K},\, dx).$$

Lemma 4.7. Let K_1/K and α be as above, and let ψ be a nontrivial additive character of K. There exists at most one α, ψ-theory of constants for K_1/K. For one to exist, it is necessary and sufficient that, for every

$$v \in \mathrm{Ker}(R_0^+(G) \longrightarrow R(G)),$$

one have

$$\varepsilon(v \cdot \alpha, \psi) = 1.$$

Let $H \subset G$, L, and V be as above. By 1.5, there exists $v \in R_0^+(H)$ whose image in $R(H)$ is $V - \dim(V)[1]$. Necessarily

$$\varepsilon_{\alpha,\psi}(V, dx) = \varepsilon(v \cdot (\alpha \circ N_{L/K}),\, \psi \circ \mathrm{Tr}_{L/K})\, \varepsilon(\alpha \circ N_{L/K},\, \psi \circ \mathrm{Tr}_{L/K},\, dx)^{\dim V}, \tag{4.7.1}$$

which gives uniqueness. The hypothesis ensures that $\varepsilon_{\alpha,\psi}$, defined by this formula, is independent of the choice of v; one easily checks axioms (0) to (4).